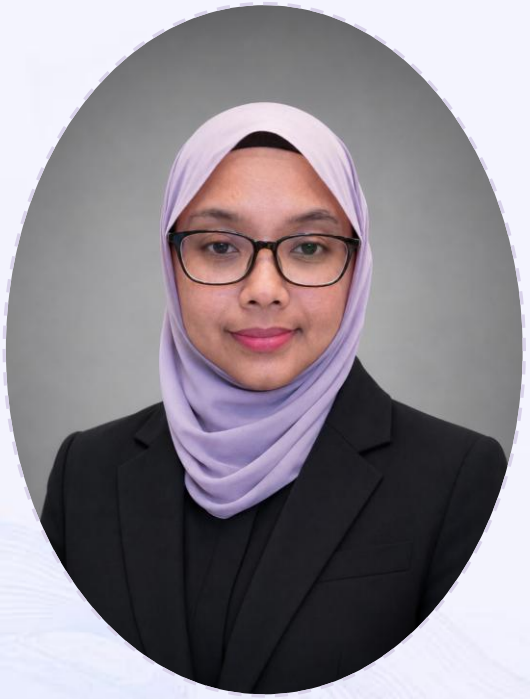
ASEAN
GEOAI FUSION 2026

Industry Applications of GeoAI: Driving Innovation with Geospatial Intelligence

[NURUL AINA ABDUL AZIZ] | [RESEARCH OFFICER, MALAYSIAN SPACE AGENCY]

[7 July 2026] • [Bootcamp 1/ Special Industry Lecturer] • ASEAN GeoAI Fusion 2026

About the Trainer



MRS. NURUL AINA ABDUL AZIZ

[RESEARCH OFFICER, MALAYSIAN SPACE AGENCY]

Bachelor of Science (Honours) in Geography from the University of Malaya with professional experience at MYSA since 2013. Work focuses on space technology, remote sensing, GIS, and artificial intelligence (AI), particularly in applying deep learning in geospatial data analysis focusing on forest fire monitoring, land use mapping, and agricultural monitoring. Research is presented at international conferences and receives several awards best paper award including the Gold Medal at the WEBCON Contest (ACRS 2023) and the Excellence in Agriculture Award at the Geospatial World Excellence Awards 2023

Application Remote Sensing | Geospatial | Deep Learning

Session Agenda

GeoAI adoption as an operational capability that converts Earth observation data into actionable intelligence for national agencies and industries

01	**Introduction: MYSA's Role in the National Space Ecosystem**	[5 minutes]
02	**Why GeoAI Matters in Earth Observation?**	[5 minutes
03	**MYSA's Operational Application Systems for Industrial Solutions**	[10 minutes]
04	**Way Forward: Advancing GeoAI at MYSA**	[5 minutes]
05	**Q&A**	[5 minutes]

MYSA'S ROLE IN THE NATIONAL SPACE ECOSYSTEM

MALAYSIAN SPACE AGENCY (MYSA)

BACKGROUND

The Malaysian Government through the Cebinet Meeting on 20 February 2018 has approved the marger of the Malaysian Remote Sensing Agency (ARSM) and the National Space Agency (ANGKASA) to form the Malaysian Space Agency (MYSA). This is one of the government's initial steps in implementing the National Space Policy 2020 where the first thrust is to strengthen government governance in the space sector.

Previously, ANGKASA focused on upstream space technology while ARSM focused on downstream. The merger of these two agencies will optimise the use of existing resources and facilities at ANGKASA and ARSM and at the same time increase the efficiency of government service delivery.

In line with the second thrust of the National Space Policy, the Malaysian Space Agency will focus on technology, infrastructure and space applications that are strategic and significant to the country, i.e. areas that have a direct impact on the economy, social, security and national sovereignty.

MISION

Leading and empowering the nation's space technology ecosystem as a whole through research and development, knowledge generation, human capital development as well as providing optimal access to space technology and infrastructure.

FUNCTION

MYSA

- LEADING THE NATIONAL SPACE POLICY implementation;
- LEADING RESEARCH AND DEVELOPMENT in Technology, application and Space Science;
- COORDINATE SATELLITE DATA
- PROVIDE SPACE SYSTEM testing and measurement services
- IMPLEMENT AND COORDINATE space exploration activities; and
- STRENGTHEN INTERNATIONAL cooperation in space field.

277 STAFFING **139** RESEARCHER

VISION

Ensuring the space technology ecosystem towards generating the economy, enhancing competitiveness, people's well-being and safeguarding the sovereignty of the country

GOAL

Leader in Research and Development in the Field of National Space Science and Technology

MYSA

MYSA

MALAYSIAN SPACE AGENCY





MYSA Headquarters
Kuala Lumpur





Space Technology Complex
Banting, Selangor



Space Operation Complex
Temerloh, Pahang

Langkawi National Observatory
Langkawi, Kedah

OPERATIONALISATION OF SPACE TECHNOLOGY IN MALAYSIA



APPLICATION DEVELOPMENT

i. REMOTE SENSING
ii. METEOROLOGY
iii. GLOBAL POSITIONING SYSTEM AND NAVIGATION
iv. COMMUNICATION
v.. SPACE SCIENCE

MISSION CONTROL STATION & DATA RECEIVING

i. DEVELOPMENT OF SYSTEM OPERATION INFRASTRUCTURE AND GROUND CONTROL STATION
ii. DEVELOPMENT OF SYSTEM OPERATION INFRASTRUCTURE AND DATA RECEIVING STATION
iii. DEVELOPMENT AND OPERATION OF AIRCRAFT SYSTEM AND UAV



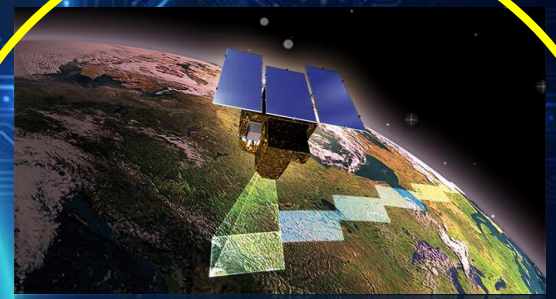
SPACE TECHNOLOGY DEVELOPMENT

i. SATELLITE DESIGN AND INTEGRATION
ii. SENSOR AND PAYLOAD DEVELOPMENT
iii. SATELLITE LAUNCHING AND LAUNCHER TECHNOLOGY
iv. EXPLORATION AND SPACE SCIENCE

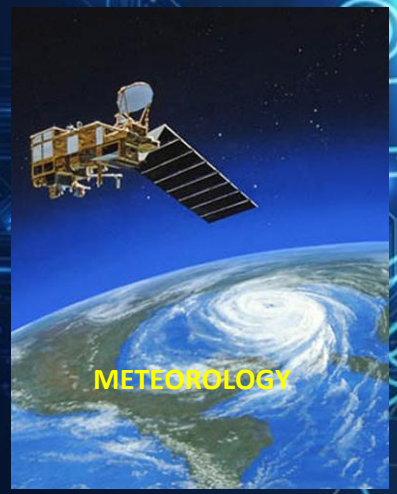
SPACE TECHNOLOGY APPLICATIONS



GLOBAL POSITIONING AND NAVIGATION



REMOTE SENSING *(EARTH OBSERVATION)*

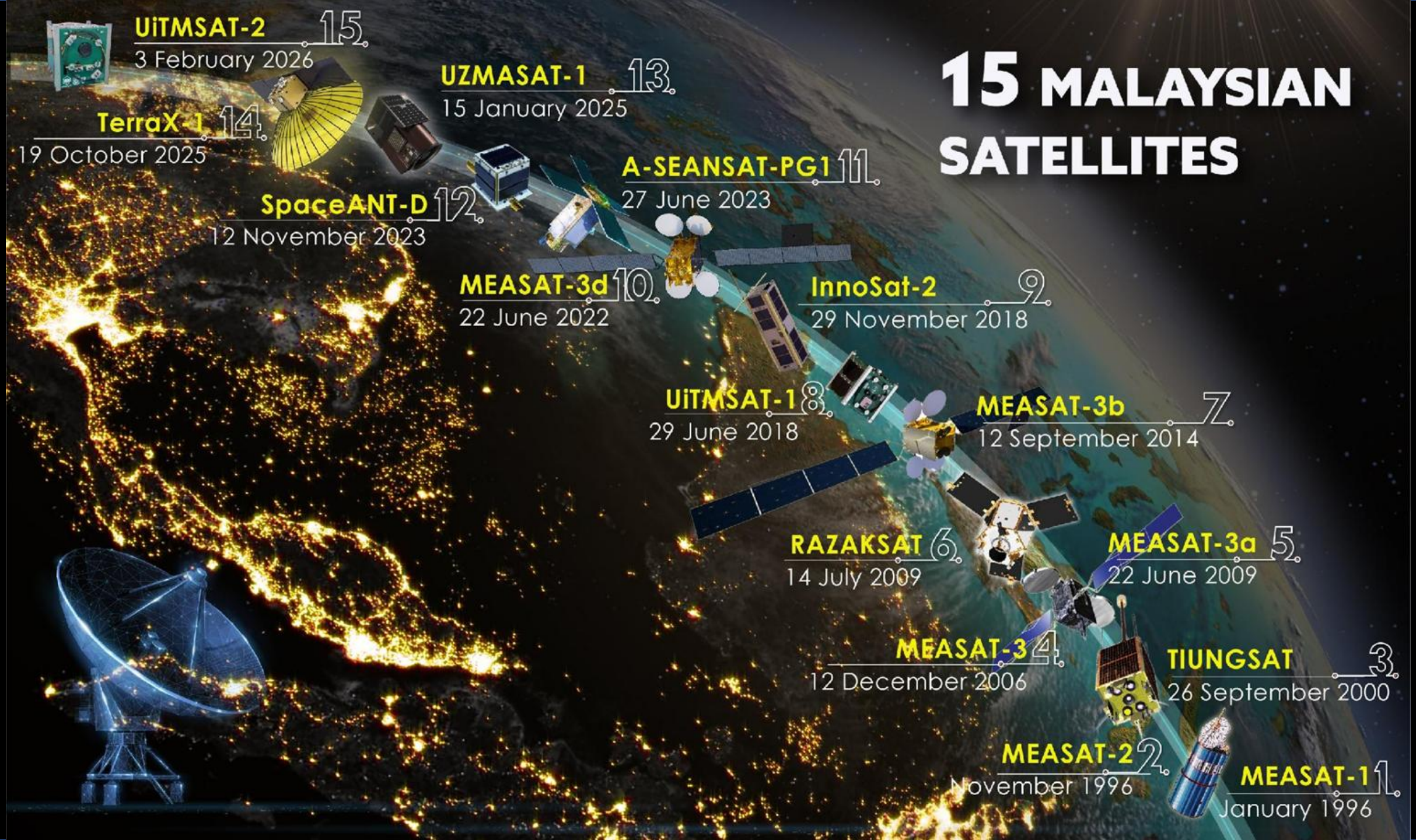


METEOROLOGY



COMMUNICATIONS

15 MALAYSIAN SATELLITES



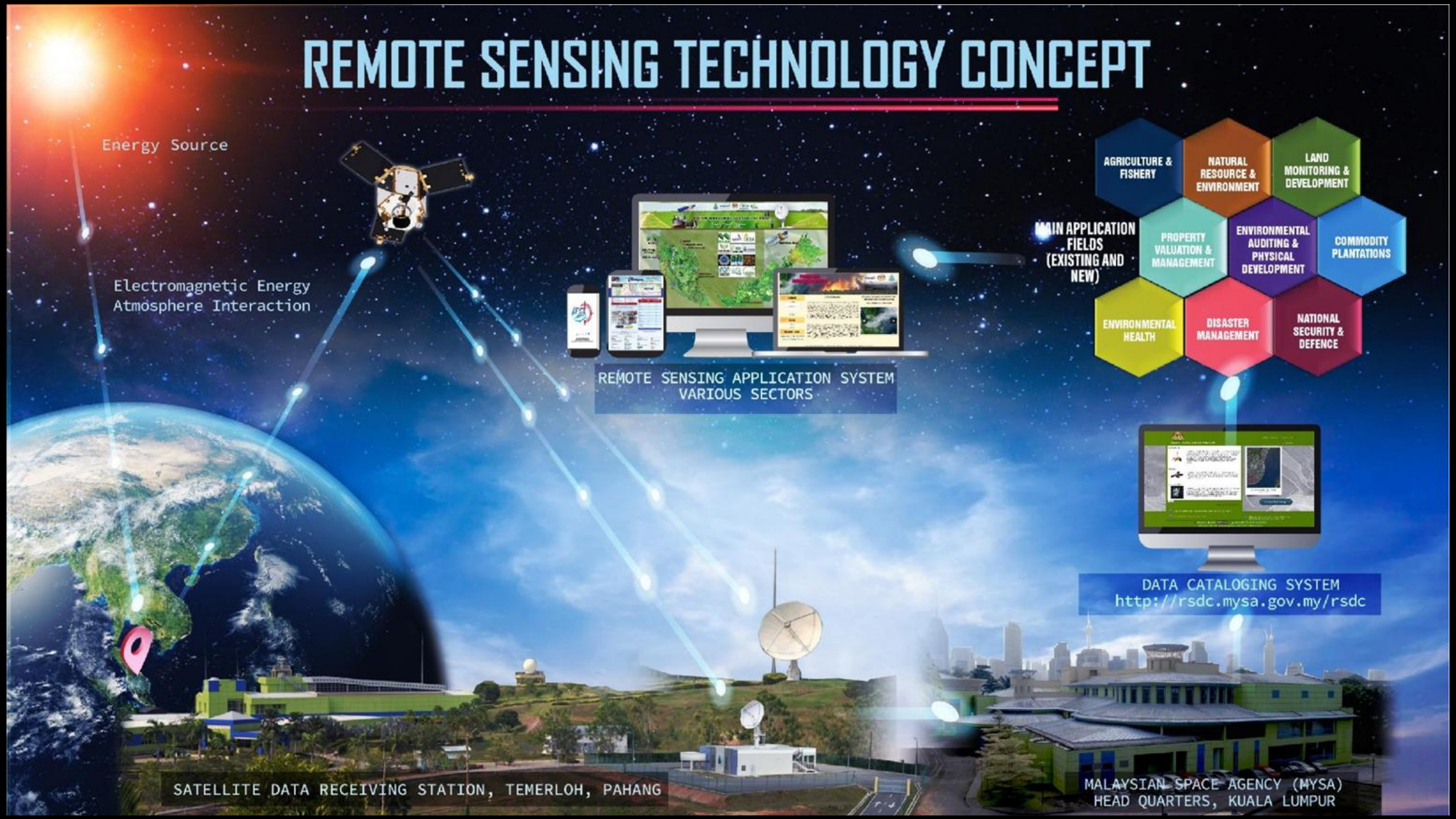
WHY GEOAI MATTERS IN EARTH OBSERVATION?

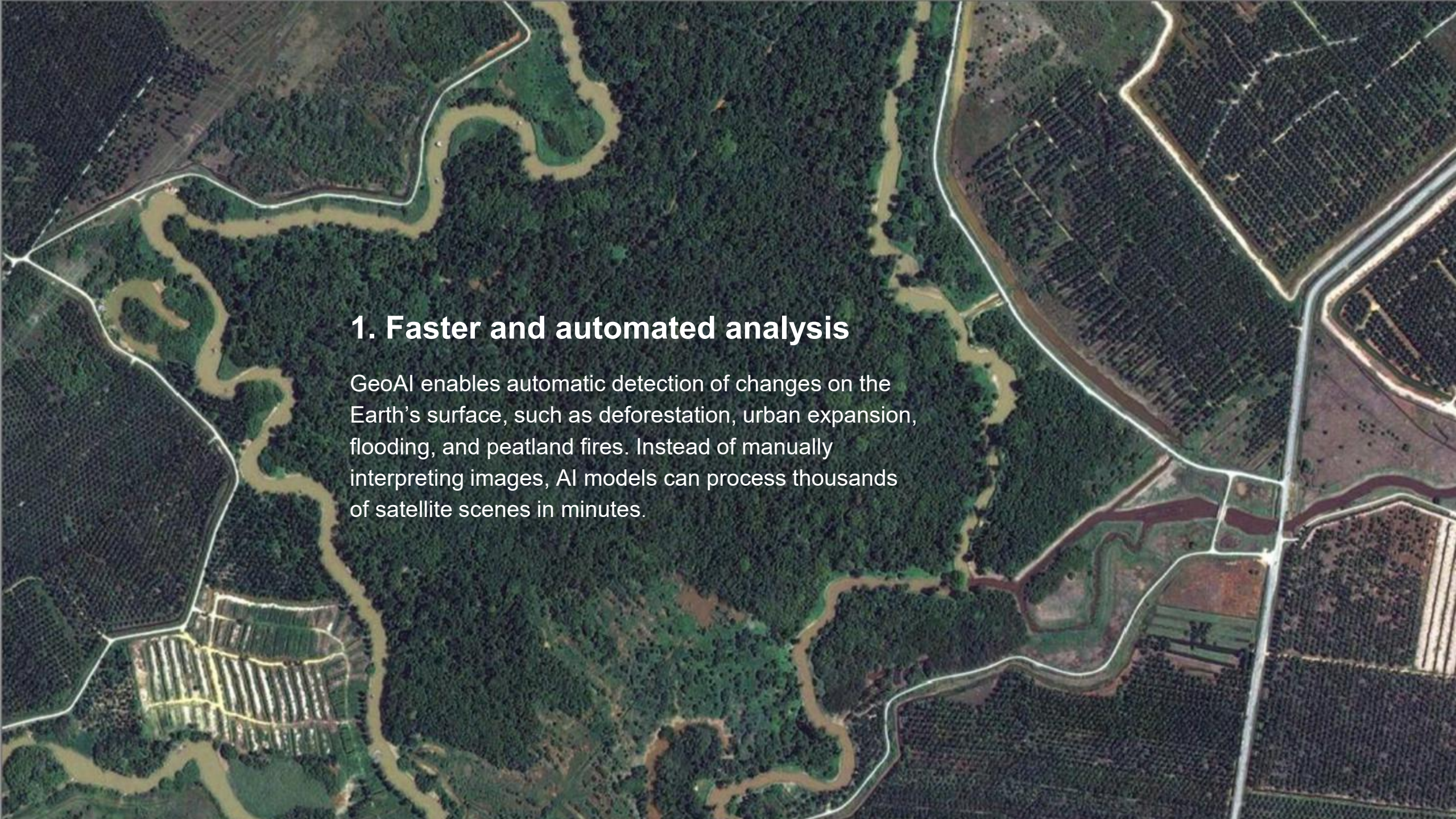
WHY GEOAI MATTERS IN EARTH OBSERVATION?

- GeoAI matters in Earth Observation because it **transforms raw satellite** and geospatial data into faster, more accurate, and more actionable insights using artificial intelligence.
- Earth Observation systems (satellites, radar, drones, LiDAR) generate **massive and continuous streams of data**.
- Traditionally, processing this data requires significant time, **manual interpretation**, and domain expertise.
- GeoAI where geospatial science is integrated with machine learning and deep learning changes this by **automating analysis and improving pattern recognition** at scale.



REMOTE SENSING TECHNOLOGY CONCEPT





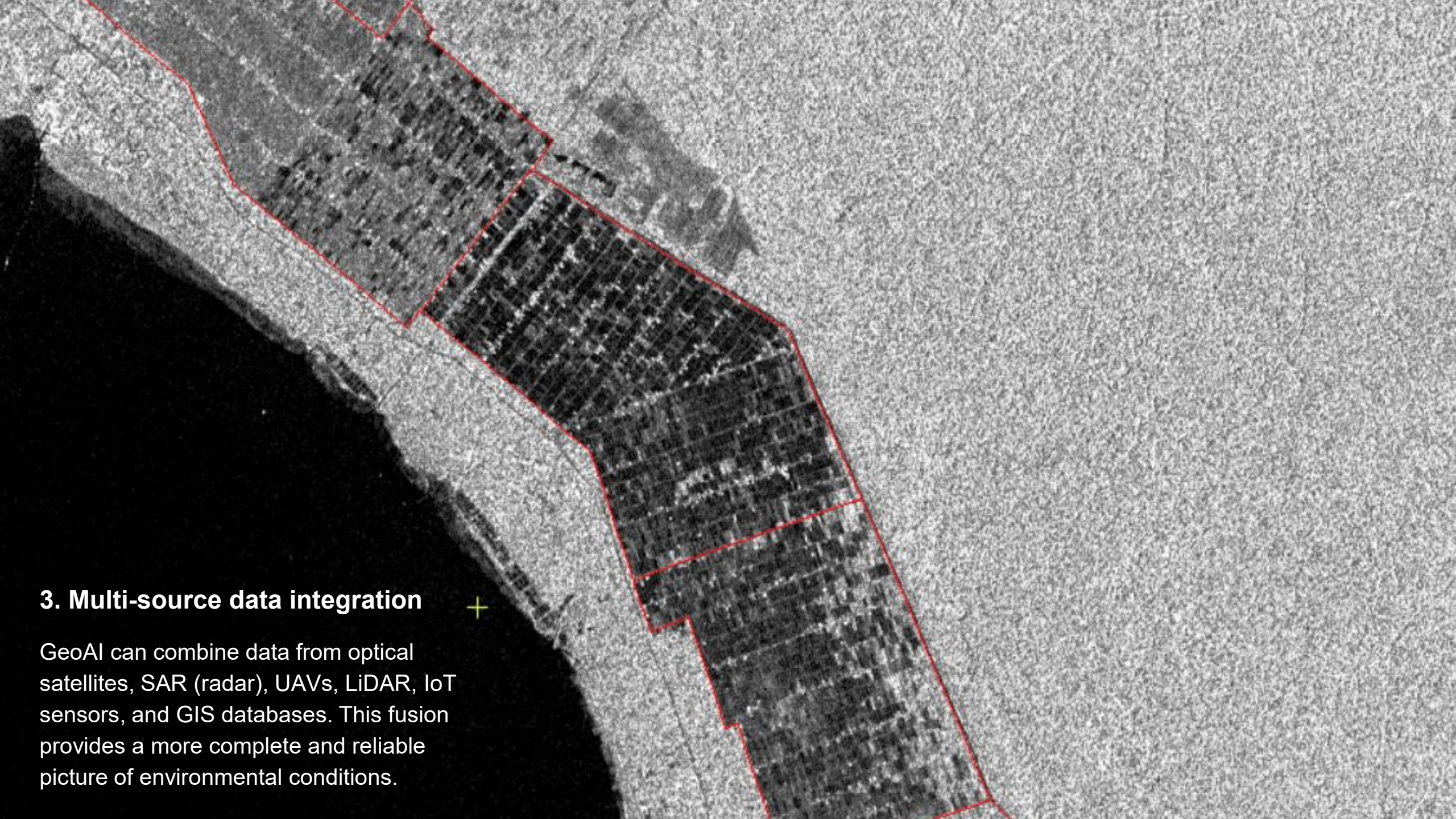
1. Faster and automated analysis

GeoAI enables automatic detection of changes on the Earth’s surface, such as deforestation, urban expansion, flooding, and peatland fires. Instead of manually interpreting images, AI models can process thousands of satellite scenes in minutes.



2. Higher accuracy and early warning capability

Machine learning models can detect subtle spatial and temporal patterns that humans may miss..



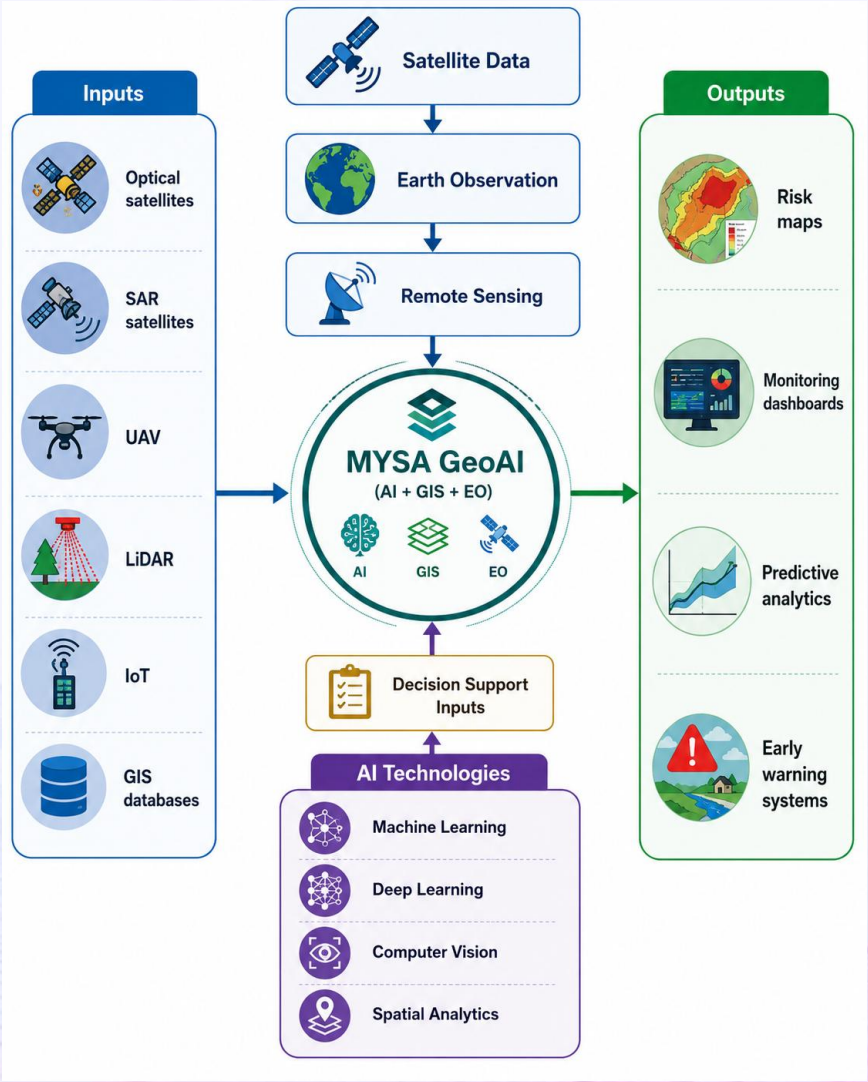
3. Multi-source data integration

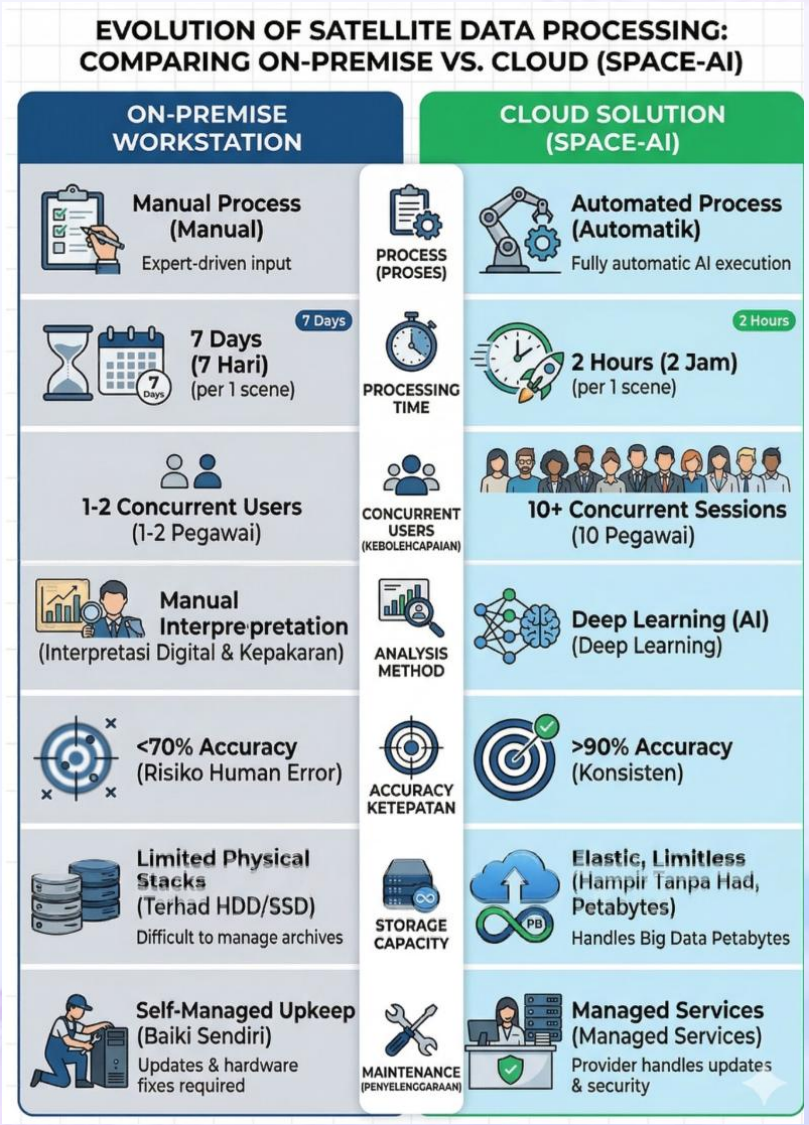
GeoAI can combine data from optical satellites, SAR (radar), UAVs, LiDAR, IoT sensors, and GIS databases. This fusion provides a more complete and reliable picture of environmental conditions.



4. Scalable decision support

GeoAI supports national-scale monitoring systems by turning Earth Observation data into decision-ready intelligence

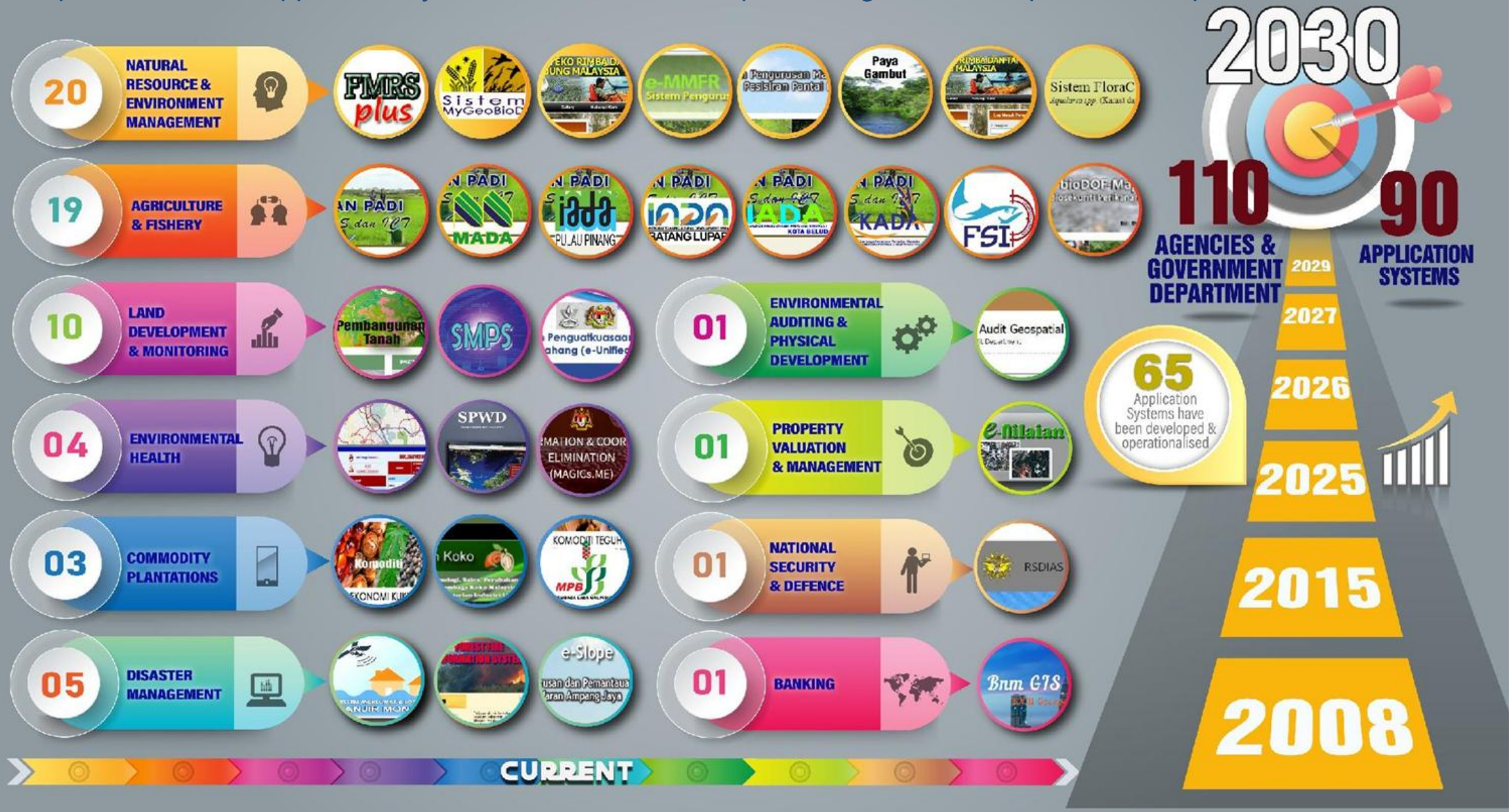

Inputs
Optical satellites
SAR satellites
UAV
LiDAR
IoT
GIS databases
Satellite Data
Earth Observation
Remote Sensing
MYSA GeoAI
(AI + GIS + EO)
AI
GIS
EO
Decision Support Inputs
AI Technologies
Machine Learning
Deep Learning
Computer Vision
Spatial Analytics
Outputs
Risk maps
Monitoring dashboards
Predictive analytics
Early warning systems


EVOLUTION OF SATELLITE DATA PROCESSING:
COMPARING ON-PREMISE VS. CLOUD (SPACE-AI)
ON-PREMISE WORKSTATION
CLOUD SOLUTION (SPACE-AI)
Manual Process (Manual)
Expert-driven input
PROCESS (PROSES)
Automated Process (Automatik)
Fully automatic AI execution
7 Days
7 Days (7 Hari)
(per 1 scene)
PROCESSING TIME
2 Hours
2 Hours (2 Jam)
(per 1 scene)
1-2 Concurrent Users
(1-2 Pegawai)
CONCURRENT USERS (KEBOLEHCAPAIAN)
10+ Concurrent Sessions
(10 Pegawai)
Manual Interpretation
(Interpretasi Digital & Kepakaran)
ANALYSIS METHOD
Deep Learning (AI)
(Deep Learning)
<70% Accuracy
(Risiko Human Error)
ACCURACY KETEPATAN
>90% Accuracy
(Konsisten)
Limited Physical Stacks
(Terhad HDD/SSD)
Difficult to manage archives
STORAGE CAPACITY
Elastic, Limitless
(Hampir Tanpa Had, Petabytes)
Handles Big Data Petabytes
Self-Managed Upkeep
(Baiki Sendiri)
Updates & hardware fixes required
MAINTENANCE (PENYELENGGARAAN)
Managed Services
(Managed Services)
Provider handles updates & security

OPERATIONAL APPLICATION SYSTEMS FOR INDUSTRIAL SOLUTIONS

APPLICATION SYSTEM DEVELOPMENT

Up untill 2025, 65 application systems have been developed through internal expertise and operationalised since 2008



PADDY MANAGEMENT SYSTEM

KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI

KEMENTERIAN PERTANIAN DAN KETERJAMINAN MAKANAN

KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI AGENSI ANGKASA MALAYSIA

JABATAN PERTANIAN

Objective

To develop, operate and improve application systems and databases based on remote sensing technology, GIS and ICT to assist agricultural agencies more efficient and effective in paddy management.

The system was fully completed in **2012** and has been in operation until now.





Monitoring paddy cultivation activities









Impact



*Monitoring paddy crops

Monitoring time ↓ **57%** on the field

Coverrage area monitored ↑ **100%** more comprehensive

government fertilizer and poison subsidies

Tepat Save **10%**

Benefit to **192,663** farmers

**Sumber Pertubuhan Peladang Kawasan (PPK) Bagan Tiang/Parit Buntar, Kerian, Perak
*Sumber Kajian Keberkesanan Pengoperasian Sistem oleh Unit Pertanian, BPICT, DOA











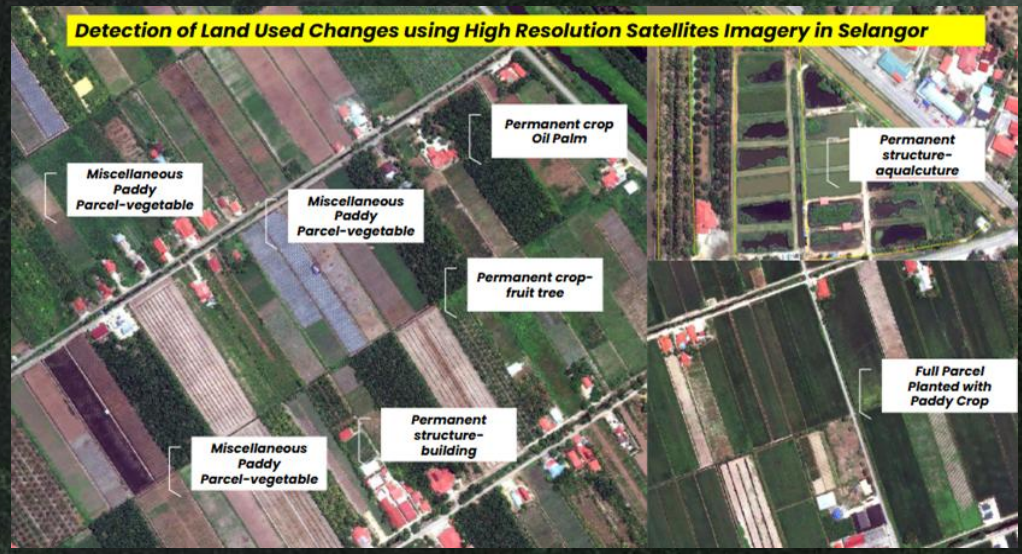








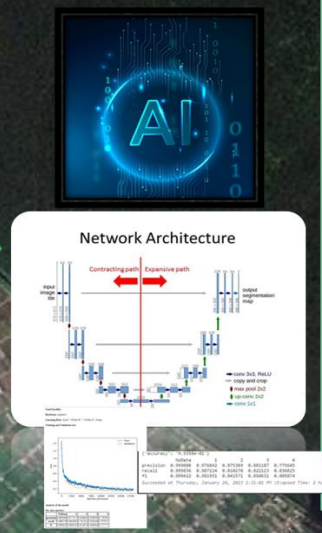
Classification of Paddy Cultivation Areas



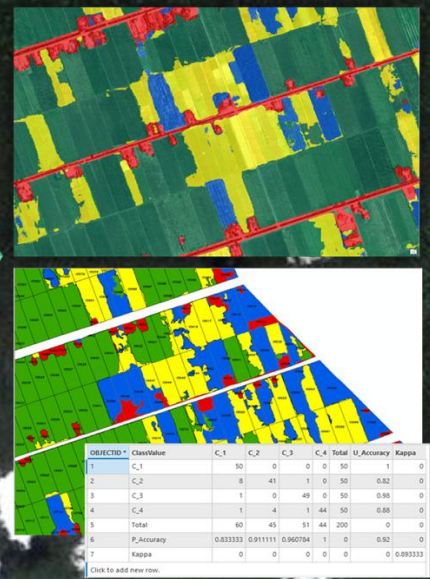
DATASET DEVELOPMENT



TRAINING OF U-NET SEGMENTATION MODUL



PREDICTED OUTPUT & EVALUATION



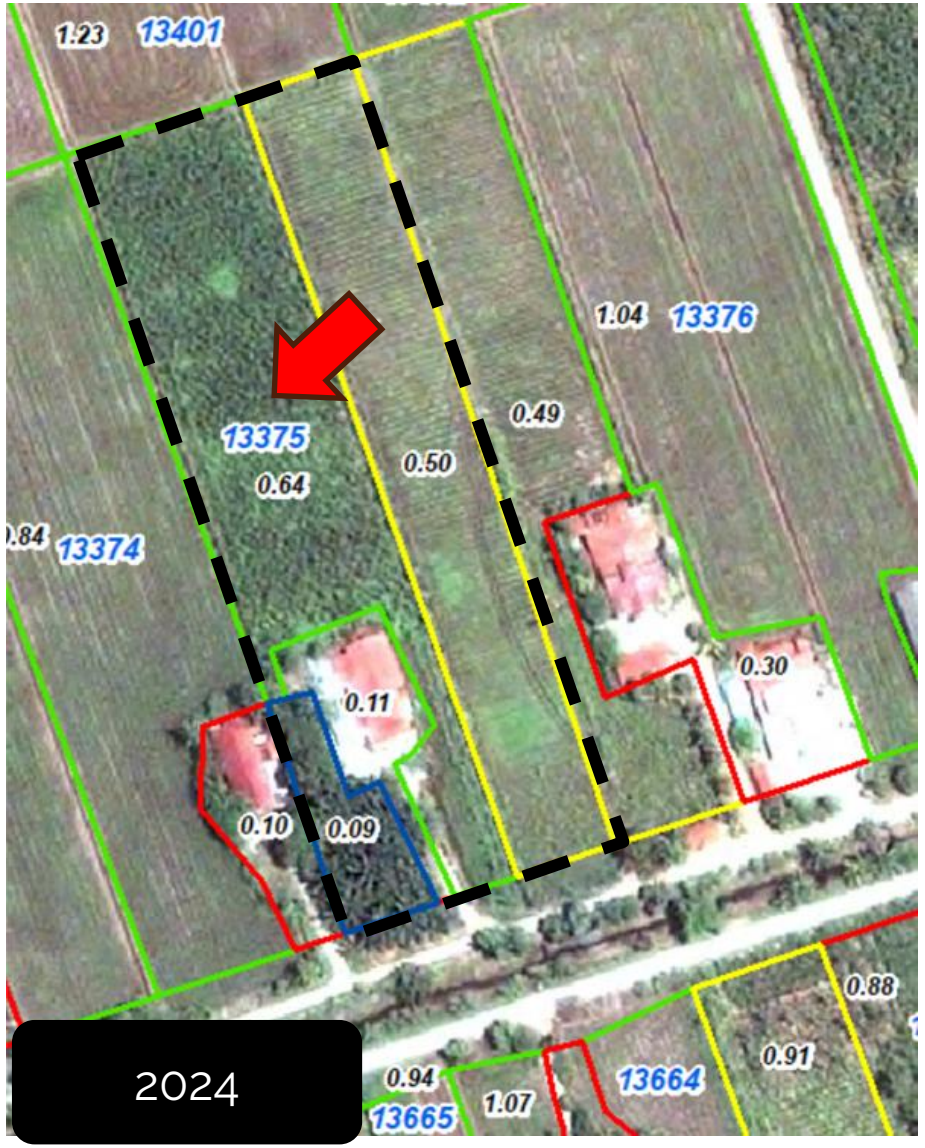
OBJECTID *	ClassValue	C_1	C_2	C_3	C_4	Total	U_Accuracy	Kappa
1	C_1	50	0	0	0	50	1	0
2	C_2	8	41	1	0	50	0.82	0
3	C_3	1	0	49	0	50	0.98	0
4	C_4	1	4	1	44	50	0.88	0
5	Total	60	45	51	44	200	0	0
6	P_Accuracy	0.833333	0.911111	0.960784	1	0	0.92	0
7	Kappa	0	0	0	0	0	0	0.893333

Click to add new row.

Segmentation of Very High-Resolution Paddy Parcels Imagery Using Deep Learning (U-Net)



2023



2024

Complexity of land cover classes in one plot (eg. Lot 13375)

Rapid land use changes for each season

Change in Actual Planted area

Previous Method: Manual Annotation of Updating Class of Paddy Field

- active paddy parcel (PA)
- miscellaneous paddy parcel (PP)
- permanent structures (SK)
- permanent crop (TK)

PLANTATION AND COMMODITIES INDUSTRY MANAGEMENT SYSTEM

KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI
KEMENTERIAN PERLADANGAN DAN KOMODITI
KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI AGENSI ANGKASA MALAYSIA
MTIB
MPOB
MPB

Objective

To develop a database and land use mapping to assist in the monitoring and management process, obtain accurate topography of commodity plantation areas and as a mechanism for monitoring, verification and assisstance and funds evaluation that have been channelled by the government for the commodity plantation sector.

The system was fully completed in **2017** and has been in operation until now.





Impact

Contribute to:

- Efficiency of monitoring and verification of large-scale plantation areas and evaluation of government subsidy
- Increase in national commodity revenue







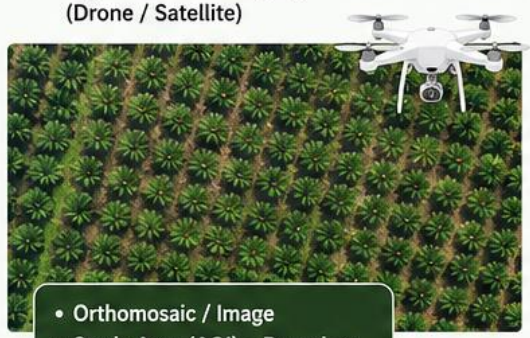


GIS AUTOMATED SOFTWARE – OIL PALM TREE COUNTING

From Imagery to Accurate Tree Count in GIS

1 DATA INPUT

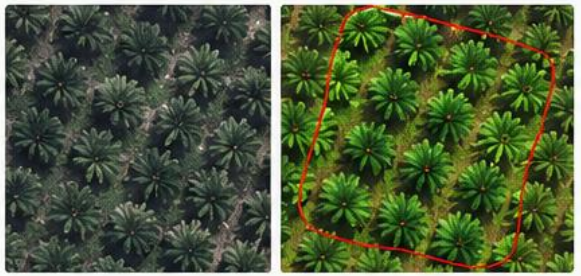
High resolution imagery
(Drone / Satellite)



- Orthomosaic / Image
- Study Area (AOI) – Boundary
- Spatial Reference

2 PREPROCESSING

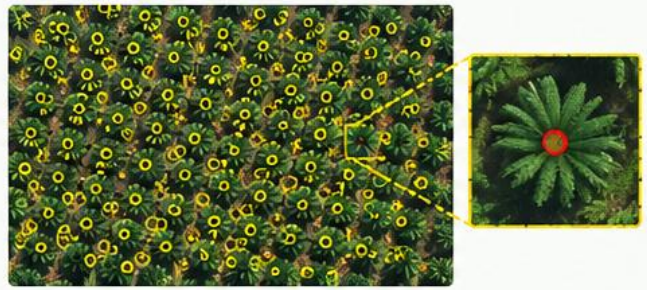
- Image enhancement
- Radiometric correction
- Clipping to AOI



Original Image | Enhanced & Clipped

3 AUTOMATED TREE DETECTION

- AI / Deep Learning model identifies palm crowns
- Detects the center (point) of each tree



Detected Tree Points (Centers)

4 COUNTING & ATTRIBUTION

- Each detected point = 1 tree
- Attributes can be added (Block, Age, etc.)

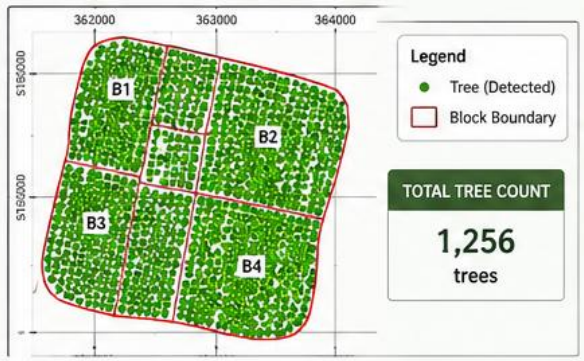
ID	Easting	Northing	Block	Age
1	362456.12	512345.67	B1	3
2	362458.95	512349.21	B1	3
3	362462.10	512352.98	B1	3
...	...	...	...	...
1256	362789.45	512789.32	B1	3

BENEFITS

- ✓ Fast & Automated
- ✓ Accurate & Consistent
- ✓ Handle Large Areas
- ✓ Supports Better Decision Making

5 SPATIAL OUTPUT IN GIS

- Tree points displayed in GIS
- Count summarized by block/zone



6 REPORT & ANALYSIS

- Automated report
- Summary by block / estate
- Export to Excel / PDF / Dashboard

OIL PALM TREE COUNT REPORT

Summary

Total Trees 1,256

Block	Tree Count
B1	1,256
B2	1,184
B3	1,098
B4	1,153
Total	4,691

Excel

PDF



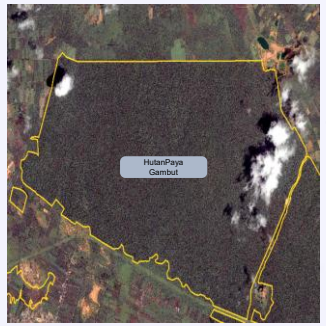
Dashboard

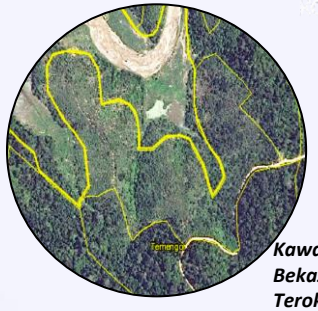
GIS + AI = Accurate Oil Palm Tree Counting

ASEAN
GEOAI FUSION 2026

Detection of open area in forest

SAMPLE OF DEFORESTATION





Kawasan Bekas Teroka Orang Asli





Tapak kongsi dan matau lama



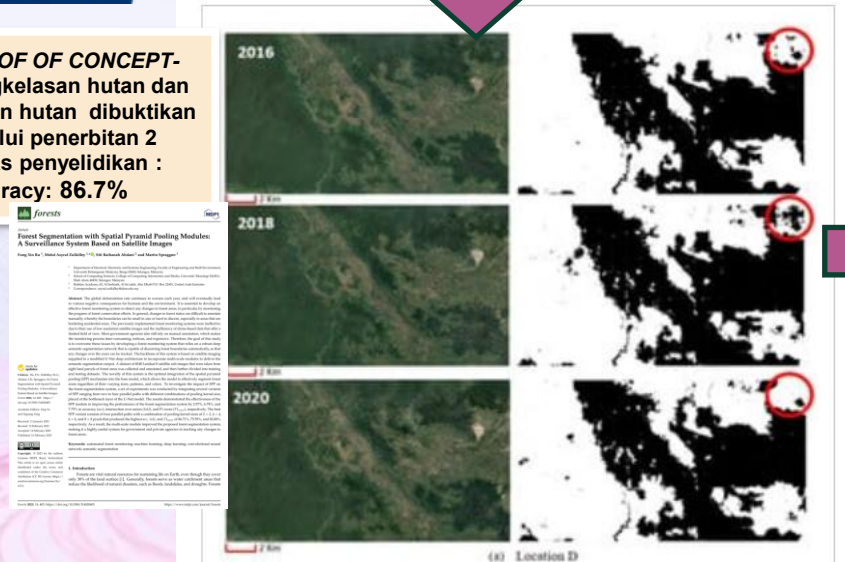


Kawasan Bekas Lombong

AI MODUL FOR FOREST CLASSIFICATION



PROOF OF CONCEPT-
Pengkelasan hutan dan bukan hutan dibuktikan melalui penerbitan 2 kertas penyelidikan :
Accuracy: 86.7%







PENGUATKUASAAN DI LAPANGAN

- Ibu Pejabat JPSM
- Jabatan Perhutanan Negeri
- Jabatan Perhutanan Daerah

SYSTEM INTEGRATION



FOREST MONITORING SYSTEM

KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI

KEMENTERIAN SUMBER ASLI DAN KELESTARIAN ALAM

KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI AGENSI ANGKASA MALAYSIA

Objective

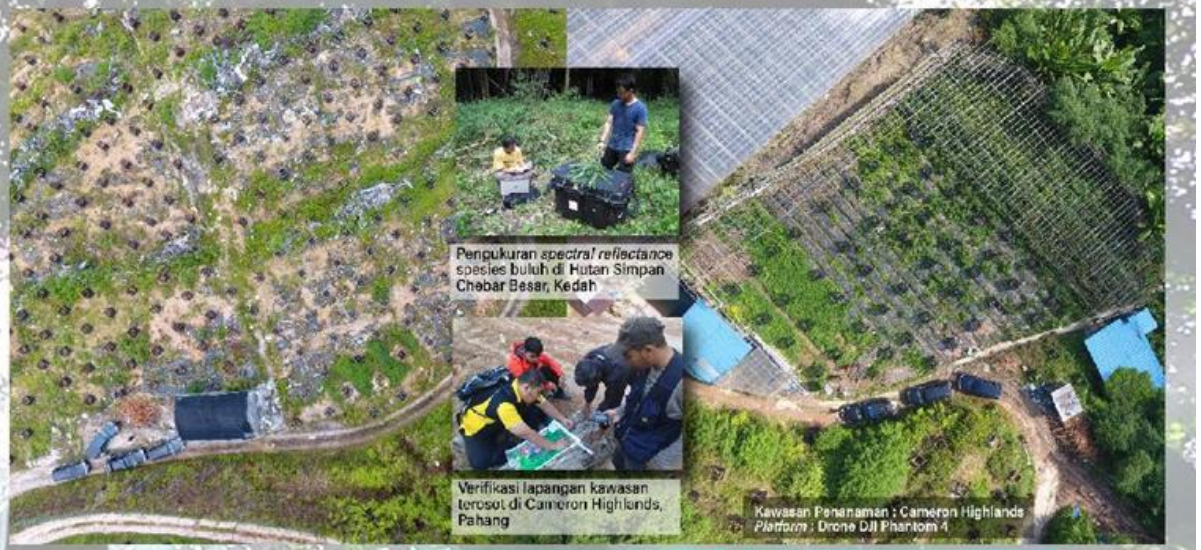
To assist the Forestry Department of Peninsular Malaysia (JPSM) to monitor forest changes using satellite images, especially in the Permanent Forest Reserve area so that it can be managed sustainably for the conservation of the country's forest treasures.

The system was fully completed in **2012** and has been in operation until now.





Siri imej bagi kawasan terbakar di Hutan Simpan Pekan.



Impact





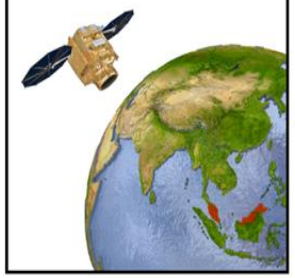
Sumber data: Jabatan Perhutanan Semenanjung Malaysia (JPSM)



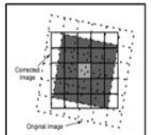
Detection of Illegal Waste Dumping via Satellite Imagery using U-Net and Open-Source Tools

Methodology

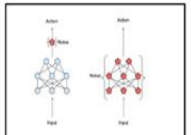
Data Collection



Preprocessing



Geometric Correction



Gaussian Noise Injection

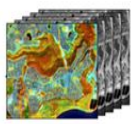
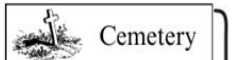


Image Enchancement

Ancillary Data



Cemetery



Forest



State Boundary

MYSA & OSM

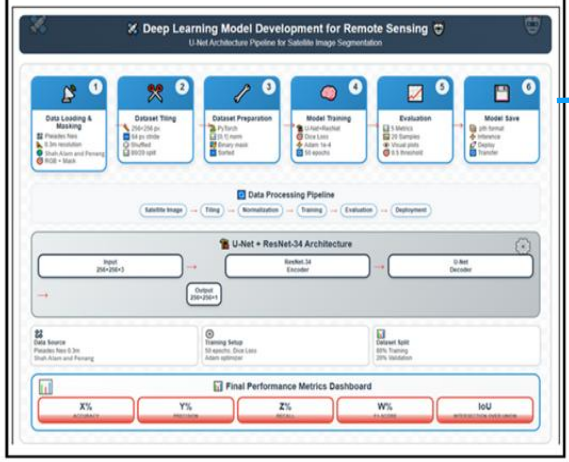
Data Annotation and Preparation



Data Augmentation

Quality check:
10% samples, IoU > 0.85

Deep Learning Model Development

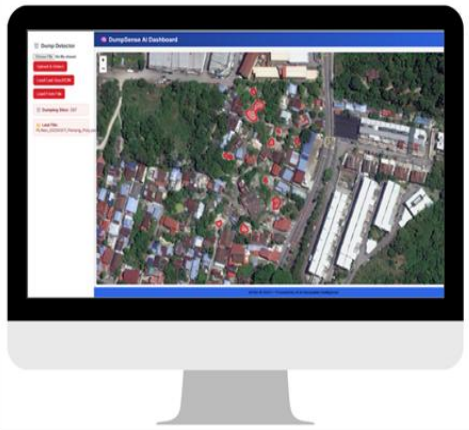


Dataset Tiling:
264 px stride, 80/20 split

Model Training:
ResNet-34, Dice Loss, Adam 1e-4, 50 epochs

Evaluation:
20 samples

Deployment and Post-Processing



Accuracy Assessment

Quantitative
- Confusion Matrix
- Model Prediction VS Ground Truth Data
- Performance Metrics

Qualitative
- Visual Interpretation
- Field Verification



- Cerapan Imej UAS MYSA
- 27 September 2019
- Rawang, Selangor



- Cerapan Imej UAS MYSA
- 21 Februari 2022
- Temerloh, Pahang

FISH SITE IDENTIFICATION SYSTEM (FSI)

KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI

KEMENTERIAN PERTANIAN DAN KETERJAMINAN MAKANAN

KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI AGENSI ANGKASA MALAYSIA

MET Malaysia

Jabatan Perikanan

LKIM

NEKMAT

Objective

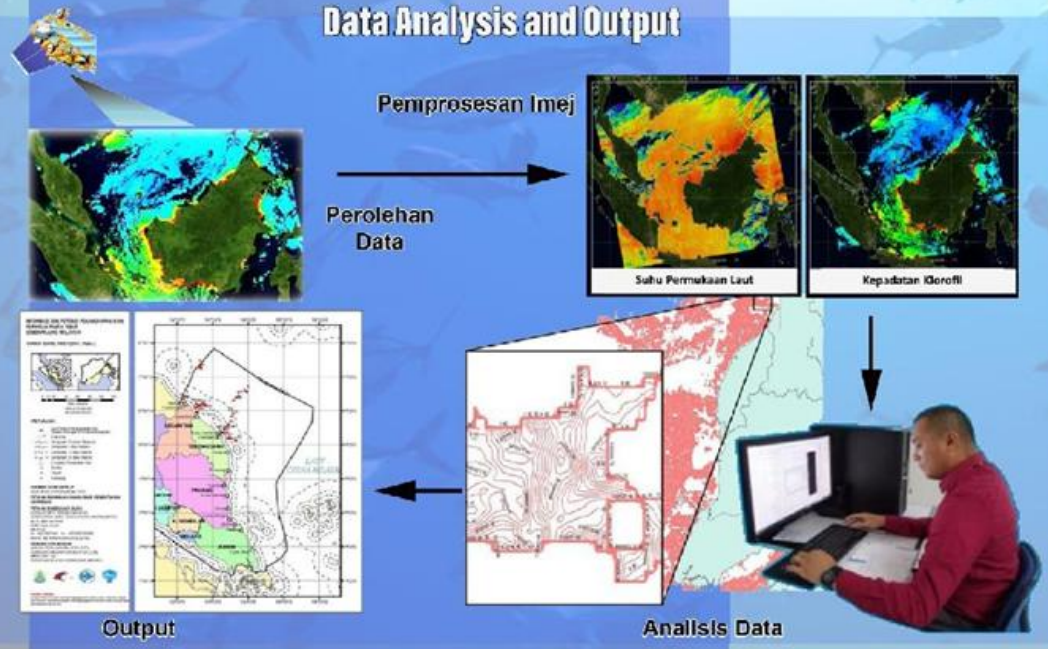
Enable the national fisheries agency to increase the income and living standards of fishermen through the use of remote sensing technology, Geographic Information System (GIS) and ICT.

The system was fully completed in **2010** and has been in operation until now.

Portable MyFSI Application



Data Analysis and Output



Impact

Memberi manfaat kepada:

1257 vesel pukat jerut

751 vesel pukat hanyut

30,161 orang nelayan di seluruh Malaysia



Increase fisherman's catches **30-50%**

Savings to the country after a 40% reduction in valuable fish imports **RM354 million**

Increase in national pelagic fish landing yield by 24% equal to **56,190** matric tone

Sumber data: Data pendaratan dan import ikan LKIM, 2014



Way Forward

WAY FORWARD
SPACE SECTOR

2030

IMPACT

1% @ RM10B

TO 5000

RM40 MILLION
A YEAR

GDP

JOB Creation

Market Expectation Data

TARGET

TOP 3 IN SOUTH EAST ASIA IN SPACE TECHNOLOGY RESEARCH AND DEVELOPMENT

50% SELF-RELIANCE IN SPACE TECHNOLOGY

ACTION

STRENGTHENING THE SPACE TECHNOLOGY ECOSYSTEM

- RESEARCH AND DEVELOPMENT IN SPACE TECHNOLOGY;
- AIT SERVICES, TT&C GROUND STATION AND DOWNLINK;
- SPACE APPLICATION DEVELOPMENT AND SERVICES; AND
- NATIONAL SATELLITE DEVELOPMENT.

MALAYSIA

Thank You

Questions & Discussion

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